Leonidas Zettas

Electrical & Computer Engineer · Power & Energy

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Graduate of the 5-year Integrated Master in Electrical & Computer Engineering (energy track), University of Western Macedonia, 2026. This portfolio presents my university project work in MV distribution automation, electrical installation design, PLC control and power electronics for renewables.

1. Diploma thesis 2. Industrial plant design 3. Apartment installation 4. PLC & control 5. Power electronics
6. Building energy study

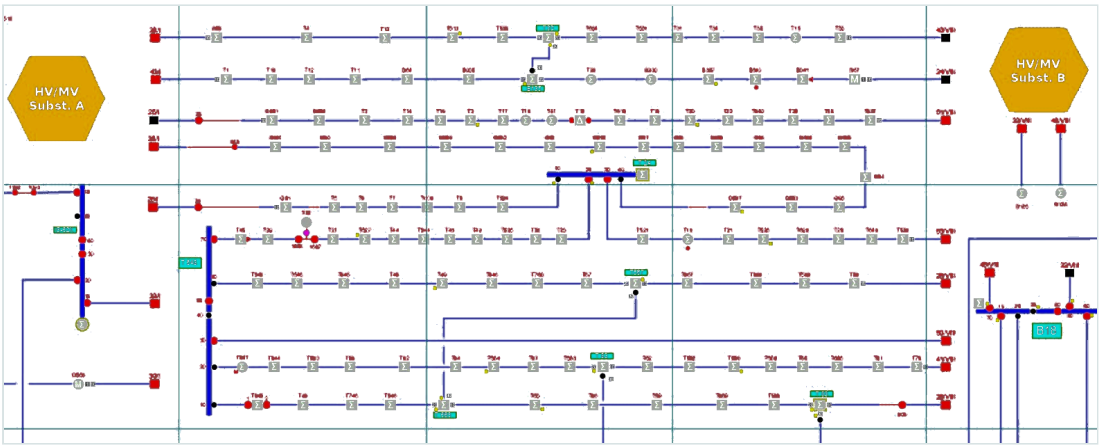
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1. Automated Switching Orders for MV Distribution Networks

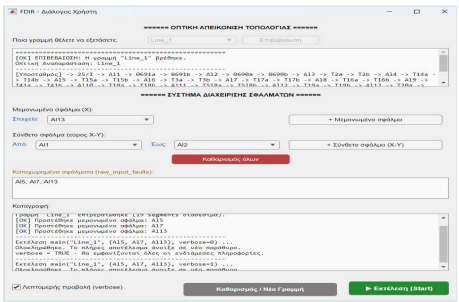
Diploma Thesis · 2026 Individual

- Developed a MATLAB application with a graphical interface that automatically writes the full switching sequence (operation note) for scheduled outages on a medium-voltage underground distribution network.
- The algorithm classifies each line as stem or branch, isolates the requested sections with the nearest switching devices, adds the CHECK / GROUND safety steps and lists the affected consumers.
- For de-energised sections it searches for alternative supply through neighbouring lines and busbars, and accepts a path only if every breaker in the chain has enough spare capacity.

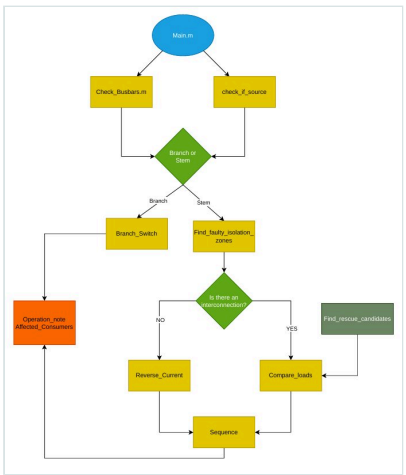
Tools: MATLAB (App GUI), Excel network database built from AutoCAD single-line diagrams · **Code:** github.com/zettasleonidas/fdir-switching-orders



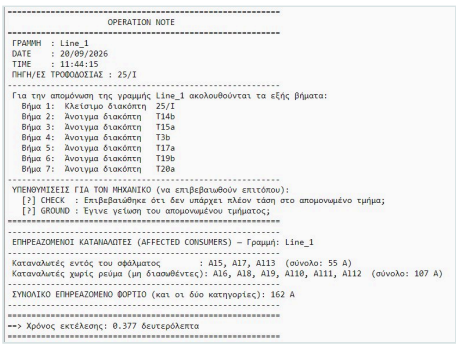
Part of the MV single-line diagram used as the study network (AutoCAD); substation names hidden.



Application interface (Greek): line selection and outage input.



Algorithm flowchart.



```
                    OPERATION NOTE
------------------------------------------------------------
ΓΡΑΜΜΗ   : Line_1
DATE     : 20/09/2026
TIME     : 11:44:15
ΠΗΓΗ/ΕΣ ΤΡΟΦΟΔΟΣΙΑΣ : 25/I
------------------------------------------------------------
Για την απομόνωση της γραμμής Line_1 ακολουθούνται τα εξής βήματα:
  Βήμα 1:  Κλείσιμο διακόπτη  25/I
  Βήμα 2:  Άνοιγμα διακόπτη   Τ14b
  Βήμα 3:  Άνοιγμα διακόπτη   Τ15a
  Βήμα 4:  Άνοιγμα διακόπτη   Τ3b
  Βήμα 5:  Άνοιγμα διακόπτη   Τ17a
  Βήμα 6:  Άνοιγμα διακόπτη   Τ19b
  Βήμα 7:  Άνοιγμα διακόπτη   Τ20a
------------------------------------------------------------
ΥΠΕΝΘΥΜΙΣΕΙΣ ΓΙΑ ΤΟΝ ΜΗΧΑΝΙΚΟ (να επιβεβαιωθούν επιτόπου):
  [?] CHECK  : Επιβεβαιώθηκε ότι δεν υπάρχει πλέον τάση στο απομονωμένο τμήμα;
  [?] GROUND : Έγινε γείωση του απομονωμένου τμήματος;
============================================================
------------------------------------------------------------
ΕΠΗΡΕΑΖΟΜΕΝΟΙ ΚΑΤΑΝΑΛΩΤΕΣ (AFFECTED CONSUMERS) – Γραμμή: Line_1
------------------------------------------------------------
Καταναλωτές εντός του σφάλματος        : A15, A17, A113  (σύνολο: 55 A)
Καταναλωτές χωρίς ρεύμα (μη διασωθέντες): A16, A18, A19, A110, A111, A112  (σύνολο: 107 A)
------------------------------------------------------------
ΣΥΝΟΛΙΚΟ ΕΠΗΡΕΑΖΟΜΕΝΟ ΦΟΡΤΙΟ (και οι δύο κατηγορίες): 162 A
------------------------------------------------------------
============================================================
--> Χρόνος εκτέλεσης: 0.377 δευτερόλεπτα
============================================================
```

Generated operation note with safety steps and affected consumers.

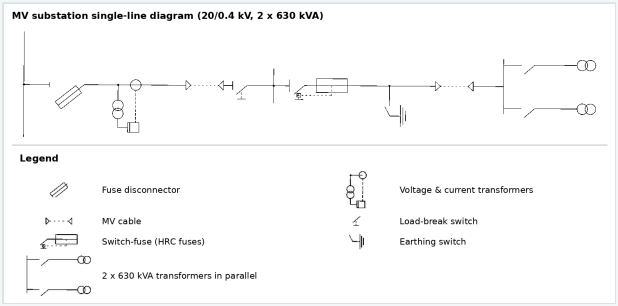
2. Electrical Design of an Industrial Plant (20/0.4 kV)

Industrial Electrical Installations · 2024–25 Team project

Tools: 4M FINE ELEC, DIALux, AutoCAD, Excel

2.1 MV/LV substation

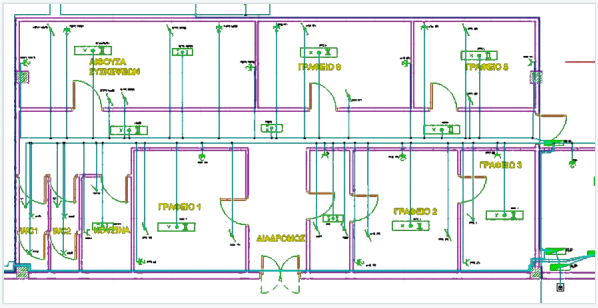
- Two 630 kVA, 20/0.4 kV transformers for parallel operation, Cu/XLPE 3×70 mm² MV cable, HRC fuse-switch protection, MV/LV switchgear and a maintenance plan.



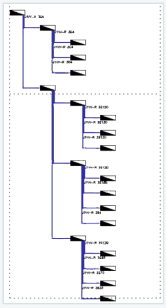
MV substation single-line diagram with legend.

2.2 LV power installation

- Plant load ≈1.08 MVA: main panel, lighting and motor sub-panels, cable sizing, protective devices and voltage-drop check for every line (below 2% for lighting and 4% for other loads).
- Chose the starting method for each machine (soft starter or variable-speed drive) and selected all equipment from manufacturer catalogues.



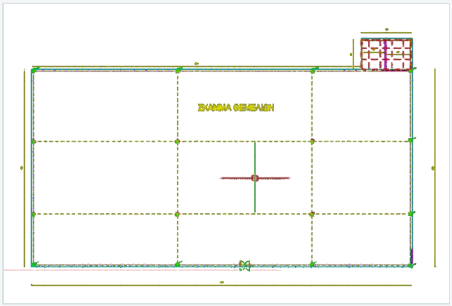
Power and lighting layout of the office area.



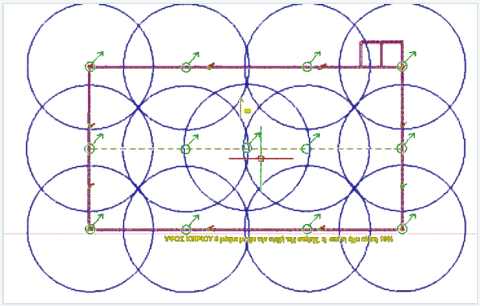
Main and sub-panel hierarchy.

2.3 Earthing and lightning protection

- Foundation earthing and lightning protection to ELOT 1197, ELOT 1424 and HD 384 (rolling-sphere method, protection level I); total earthing resistance 0.67 Ω, below the 1 Ω target.



Foundation earthing system.



Lightning protection (rolling-sphere method).

2.4 Power-factor correction and installation certificate

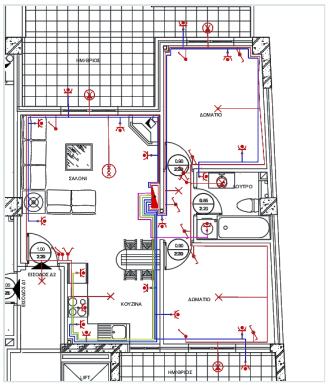
- Correction to cos φ = 0.95 (≈95 kVAr → 100 kVAr automatic capacitor bank, 7 steps), electricity-tariff comparison and payback; completed the Greek installation certificate (ΥΔΕ).

3. Electrical Installation Drawings of an Apartment

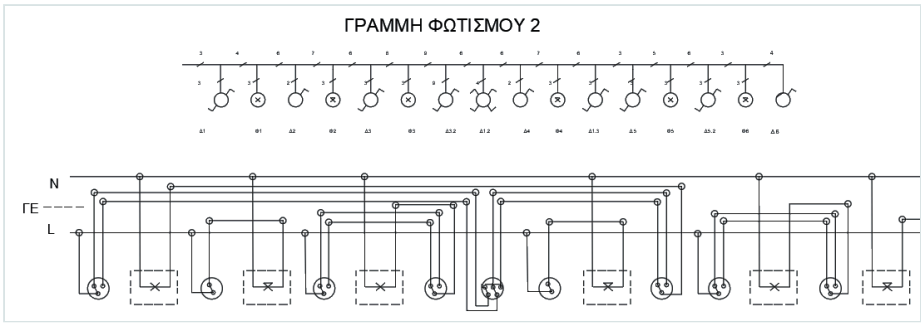
Technical Drawing (CAD) · 2020 Individual

- Drew the full electrical installation of an apartment in AutoCAD: lighting circuits, general socket circuits and dedicated lines (kitchen, fridge, washing machine) placed on the floor plan.
- Prepared the single-line diagram of every circuit and the matching multi-line wiring diagrams (L, N, PE), showing one-way, two-way and intermediate switching of the luminaires.

Tools: AutoCAD



Floor plan with circuits.



Lighting circuit 2: single-line (top) and multi-line wiring diagram (bottom).

4. PLC Programming & Motor Control

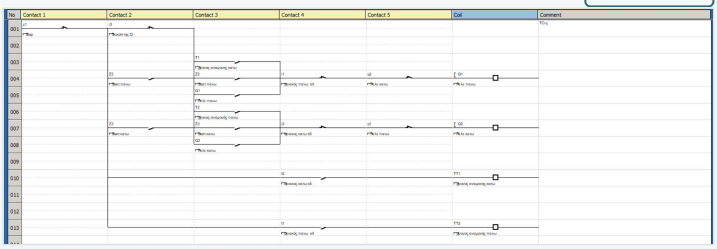
Energy Automation 2023 · Industrial Installations 2024

Tools: Zelio Soft 2 (ladder) on a Schneider Zelio Logic smart relay SR3B261FU, AutoCAD · **Programs:** github.com/zettasleonidas/plc-zelio-ladder

4.1 Skip hoist for building materials

2023 Individual

- Ladder program for a bucket hoist driven by a reversing motor: up (Q1) and down (Q2) contactor outputs with electrical interlock.
- Top and bottom limit switches stop the motor; two waiting timers hold the bucket for unloading at the top and loading at the bottom, then restart the next movement automatically. Stop button and main switch.

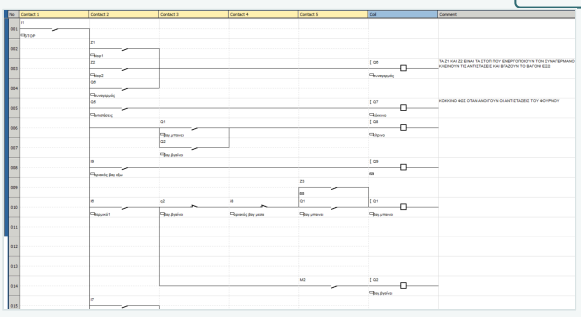


Complete ladder program.

4.2 Electric furnace with automatic trolley

2023 Individual

- Sequence: start → trolley enters → door closes only when the trolley is fully inside → heaters on → at 290 °C (thermostat) and after a timer the heaters switch off, the door opens and the trolley returns.
- Green / yellow / red status lamps; two emergency stops (inside and outside) switch off the heaters, bring the trolley out and trigger the alarm.

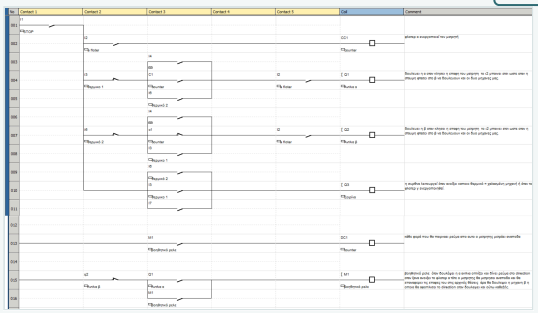


First rungs of the ladder program (38 rungs in total).

4.3 Sewage pumping station with two pumps

2023 Individual

- Float a starts one pump; if the level keeps rising, float b starts the second pump as well.
- Duty alternation with an up/down counter and an auxiliary relay, so the two pumps take turns starting first; if one pump's thermal relay trips the other runs directly. A siren sounds on an overload trip or when the high-high level float c is reached.

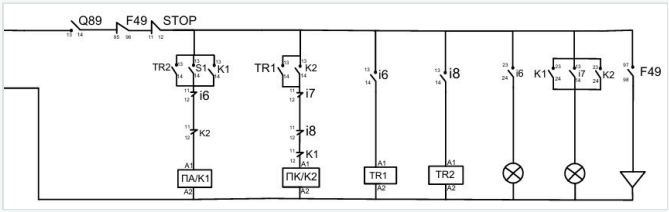


Ladder program with the counter-based pump alternation.

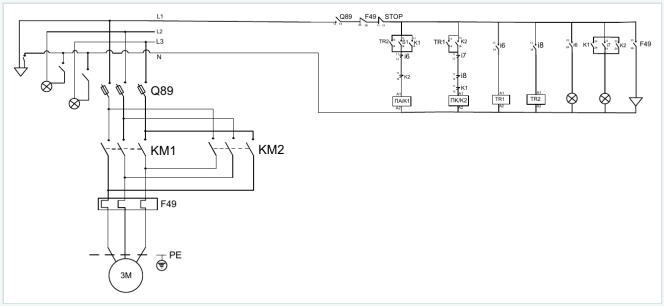
4.4 Automatic garage door: motor control circuits

2024 Team project

- Control and power circuits for an 11 kW reversing motor: interlocked contactors KM1/KM2, limit switches, auto-close timer (20 s) and obstacle timer (5 s), photocell, status lamps and overload siren.
- Sized the supply and branch circuits (cables, MCBs, RCD) and selected every component from catalogues; drew the single-line diagram and floor plan.



Control circuit: interlocks, timers TR1/TR2, photocell i7, lamps and siren.



Power circuit: fuses Q89, reversing contactors KM1/KM2, thermal relay F49, 3-phase motor.

5. Power Electronics for Wind and Solar

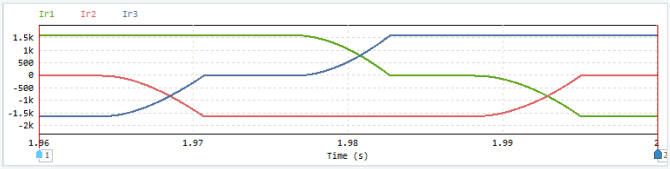
Power Electronics I, II & Special Topics · 2023–25 Team projects

Tools: PSIM, Excel

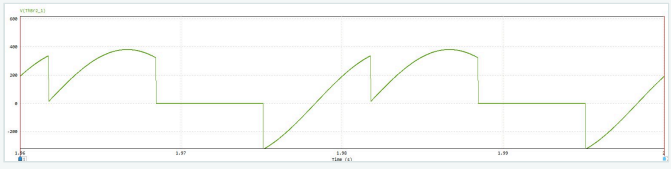
5.1 Wind turbine with doubly-fed induction generator

2023

- 2.5 MW, 690 V DFIG with slip-power recovery (diode rectifier + thyristor inverter). Calculated slip, rotor power, DC-link values and the 6n±1 current harmonics, and compared them with PSIM.



The three rotor currents (PSIM).

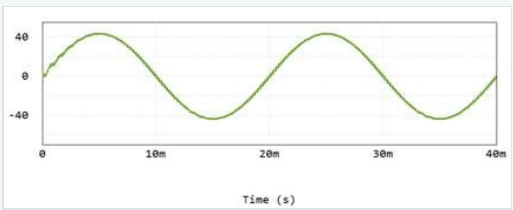


Voltage across one thyristor of the inverter (PSIM).

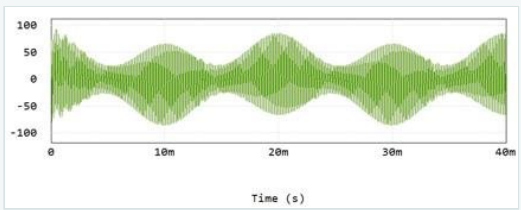
5.2 Grid-connected PV system

2024

- 4.8 kW boost converter (160–240 V → 400 V DC bus) and a 6 kVA single-phase SPWM inverter; sized and selected the components, and compared unipolar and bipolar modulation, including THD.



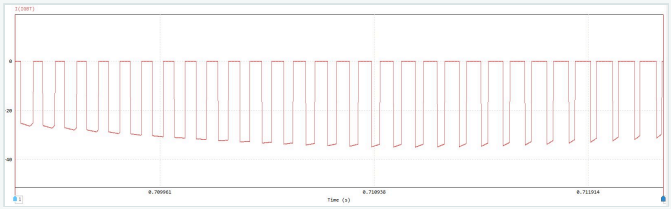
Inverter load current after the LC filter (PSIM).



Output-filter capacitor current with SPWM switching ripple (PSIM).

5.3 PV system with battery storage 2025

- Isolated flyback converter charging a 24 V battery bank (35 kHz, chosen for 97% efficiency) and a series-resonant converter with a high-frequency transformer; PSIM waveforms for each stage.



Switch current of the series-resonant converter (PSIM).

6. Energy Performance Study of a School Building

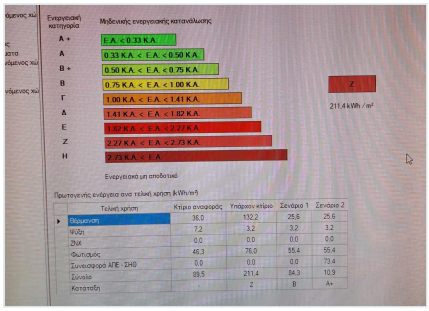
Building Energy Analysis · 2024 Team project

- Energy audit of a three-storey school (≈1,480 m², climate zone C) in TEE-KENAK, the Greek software for the EU Energy Performance of Buildings Directive: U-values, thermal losses, and the heating, cooling and lighting systems.
- Existing building rated class Z (lowest). Upgrade scenario 1 (rock-wool insulation: 12 cm on walls, 10 cm on roof; new thermal windows and doors; heat pump) → class B; scenario 2 adding a 9.9 kW rooftop PV system with net metering → class A+.

Tools: TEE-KENAK, Excel, AutoCAD



The building (image provided by the course).



TEE-KENAK results: existing building, scenario 1 and scenario 2.